FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2020

TEST DATE: 08-12-2019

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

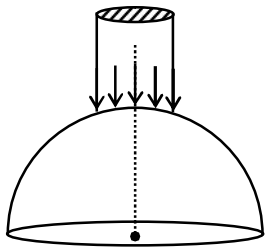
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A cylindrical beam of diameter (d = R) is incident on a glass hemisphere of radius R and refractive index $\left(\mu = \sqrt{2}\right)$ on the curved surface symmetrically as shown in the figure. The radius of circular bright spot formed on the flat surface of the hemisphere is:



(A) $\frac{R}{4\sqrt{2}}\left[2\sqrt{2}-\sqrt{7}+\sqrt{3}\right]$

(B) $\frac{R}{2\sqrt{2}}\left[2\sqrt{2}-\sqrt{7}+\sqrt{3}\right]$

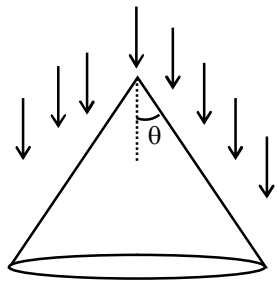
(C) $\frac{R}{4\sqrt{2}}\left[2\sqrt{2}+\sqrt{7}-\sqrt{3}\right]$

(D) $\frac{R}{10}\left[\sqrt{21}-1\right]$

2. A silicon diode requires minimum current of 2mA to be above its knee point of its (V-I) characteristics. Assuming voltage across diode is constant above its knee point. The value of maximum resistance that should be connected in series with diode and battery of 10 V such that diode operates above its knee point. (Assume resistance of diode to be zero)

(A) 5 kΩ
(B) 5.35 kΩ
(C) 4.65 kΩ
(D) 9.30 kΩ

3. Light of intensity I falls parallel to the axis on the curved surface of a perfectly reflecting right circular cone having semi vertical angle θ and base radius R as shown in figure. The value of force acting on the cone due to light is (where c is speed of light in vacuum)



(A) $\frac{\pi R^2 I}{c}(1-\cos\theta)$

(B) $\frac{\pi R^2 I}{2c}$

(C) $\frac{2\pi R^2 I}{c}$

(D) $\frac{\pi R^2 I}{c}(1-\cos 2\theta)$

4. Consider a plane progressive wave propagating along negative x-direction whose frequency is f. At $t = t_0$ the displacement associated with the wave is given by the following equation:
$y(x, t = t_0) = A \cos kx$
Displacement associated with the wave as a function of x at $t = 2t_0$ can be represented by equation.
(A) $A \cos (kx + 2\pi ft_0)$
(B) $A \sin (kx + 2\pi ft_0)$
(C) $A \cos (kx - 2\pi ft_0)$
(D) $A \cos (kx + 4\pi ft_0)$

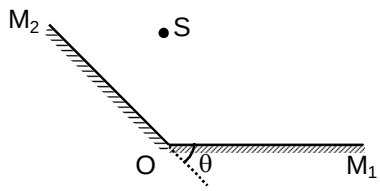
5. A sample contains two radioactive materials A and B with half life 5 hours and 2 hours respectively. The nucleus A decays into B and B decays into stable nucleus C. At t = 0 activities of both samples were equal. The ratio of activity of A to the activity of B when activity of B is maximum, is
(A) 1
(B) $\frac{5}{2}$
(C) $\frac{2}{5}$
(D) $\frac{4}{25}$

6. Water is flowing in a circular pipe of radius r and thickness t with constant speed. A valve installed in the pipe is closed almost instantaneously resulting in an abrupt rise in water pressure. What should be the value of maximum speed of the flow of water such that pipe does not crack due to abrupt rise in pressure. Given density of water ρ, speed of sound in water v and tensile strength of steel is σ.
(A) $\frac{\sigma t}{\rho v r}$
(B) $\frac{\sigma t}{2\rho v r}$
(C) $\frac{\sigma^2 t}{\rho v r^2}$
(D) $\frac{2\sigma t}{\rho v r}$

7. In a nuclear reaction $p + {}^{15}N \rightarrow {}^{15}O + n$

If proton were to collide with ^{15}N at rest and initiate above reaction. The minimum value of Kinetic energy of proton for which above reaction can take place is
Given: m(p) = 1.007825 amu, m(n) = 1.008665 amu
m(^{15}O) = 15.0031 amu, m(^{15}N)= 15.000 amu (Take 1 amu = 931.5 MeV/c^2)
(A) 3.67 MeV
(B) 3.91 MeV
(C) 3.30 MeV
(D) 3.00 MeV

8. In the arrangement shown in figure, S is a point source, M_1 and M_2 are two plane fixed mirrors inclined at an angle θ with each other. If OS = b, then the value of distance between two images formed.



(A) b sin 2θ
(B) 2b sin θ
(C) 2b sin 2θ
(D) b sin θ

9. Two optical medium of refractive index μ and 3/2 contains 3 and 2 waves of same monochromatic radiation in thickness t and 2t. The value of μ is
(A) 3
(B) 4.5
(C) 2.5
(D) 2

10. The electrons of kinetic energy 75 eV each are diffracted from a metal foil having lattice spacing d = 30Å. If the first maxima occurs when incident at an angle θ from the atomic layer, then the value of sin θ is
(A) 0.023
(B) 0.253
(C) 0.707
(D) 0.047

11. A plane progressive electromagnetic wave travelling in vacuum along +z direction has a wavelength of 6 mm. The variation in the electric field occurs in x-direction with amplitude 50 Vm^{-1}. The equation of electric field and magnetic field as a function of position and time are given by:

(A) $(50\ Vm^{-1})\cos(2\pi\times10^{11}\ s^{-1})\left(t-\frac{x}{c}\right)\hat{i}$, $(\frac{5}{3}\times10^{-6}\ T)\cos(2\pi\times10^{11}\ s^{-1})\left(t-\frac{x}{c}\right)\hat{j}$

(B) $(50\ Vm^{-1})\sin(\pi\times10^{11}\ s^{-1})\left(t-\frac{z}{c}\right)\hat{i}$, $(\frac{5}{3}\times10^{-7}\ T)\sin(\pi\times10^{11}\ s^{-1})\left(t-\frac{z}{c}\right)\hat{j}$

(C) $(50\ Vm^{-1})\sin(\pi\times10^{11}\ s^{-1})\left(t-\frac{x}{c}\right)\hat{i}$, $(6\times10^{6}\ T)\sin(\pi\times10^{11}\ s^{-1})\left(t-\frac{y}{c}\right)\hat{j}$

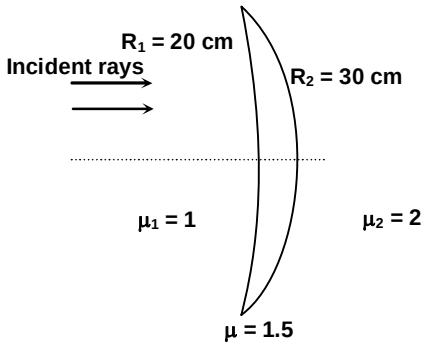
(D) $(50\ Vm^{-1})\cos(\pi\times10^{11}\ s^{-1})\left(t+\frac{z}{c}\right)\hat{i}$, $(6\times10^{6}\ T)\sin(\pi\times10^{8}\ s^{-1})\left(t+\frac{z}{c}\right)\hat{j}$

12. If x, v and a denote the displacement, velocity and acceleration of a particle executing S.H.M. of time period T and frequency ν. Then which of the following quantities does not change with time?
(A) $\frac{aT}{x}$
(B) $a^2T+4\pi\nu^2$
(C) $\frac{aT}{\nu}$
(D) none of these

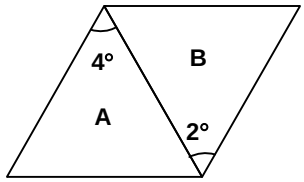
13. An equilateral prism produces an angle of deviation 60° when the angle of incidence is 60°. The angle of deviation produced when the ray is incident normal to the surface is
(A) 30°
(B) 45°
(C) 60°
(D) can't be determined

14. Magnitude of focal length of the lens shown in the figure is

(A) 240 cm
(B) 60 cm
(C) 48 cm
(D) 30 cm



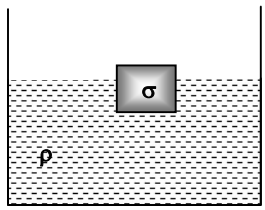
15. Two thin prisms are combined as shown in the figure. Find the total angular dispersion suffered by a white light after it is passed through the combination of two prisms. The refractive indices of prisms A and B for different lights are listed in the table below.



Color	Prism A	Prism B
Violet	1.5	1.7
Yellow	1.45	1.6
Red	1.4	1.5

(A) 0.8°
(B) 0°
(C) 3.0°
(D) 0.6°

16. A cubical block of side a, density σ is floating in a liquid of density $\rho(>\sigma)$ in equilibrium. Block is pushed down very slowly until it is completely getting inside the water. Work done by the pushing force in the process is (assume that area of cross section of container in which liquid is kept is very large as compared to a^2)



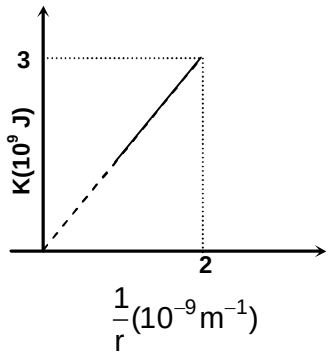
(A) $a^4g\sigma$

(B) $a^4g\left(\rho-\frac{\rho^2}{\sigma}\right)$

(C) $a^4g\left(\frac{\rho}{2}-\sigma+\frac{\sigma^2}{2\rho}\right)$

(D) $a^4g\left(\sigma-\frac{\sigma^2}{\rho}\right)$

17. A graph of kinetic energy K of an asteroid versus $\frac{1}{r}$ (where r is the distance from the centre of the planet A) is shown in the figure. Asteroid falls directly towards the centre of the planet A. What is the value of mass of the asteroid? (M is mass of planet A, G is universal gravitational constant in SI units)



(A) $\frac{3}{2MG}\times10^{18}Kg$

(B) $\frac{450}{MG}\times10^{18}Kg$

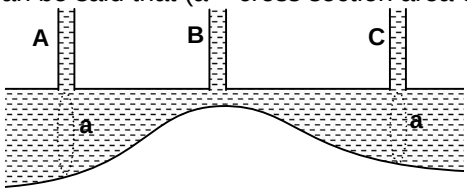
(C) $\frac{2}{MG}\times 10^{18}$Kg

(D) $\frac{600}{MG}\times 10^{18}$Kg

18. Time period of a particle executing linear S.H.M. is T = 4π sec. There is a point P at a distance x = 2 cm from the mean position O. When the particle passes through the point P towards O it has speed v = 1 cm/sec. The minimum time after which it will again return to point P is
(A) (π/4) sec
(B) 3π sec
(C) (3π/4) sec
(D) 2π sec

19. A small air bubble of radius r is found to be formed at a depth of H from open surface of liquid contained in a beaker. If T is surface tension, ρ is density of the liquid and p_0 is atmospheric pressure then pressure inside the bubble is (r = radius of bubble)
(A) $P_0+\frac{4T}{r}-\rho gH$
(B) $P_0-\rho gH+\frac{2T}{r}$
(C) $P_0+\rho gH+\frac{2T}{r}$
(D) $P_0+\frac{4T}{r}+\rho gH$

20. An ideal fluid is flowing through a horizontal pipe as shown in the figure. Three tubes A, B and C are connected to the pipe. Radii of the tubes A, B and C at junction are 2r, r and 2r respectively. It can be said that (a = cross section area of tube)

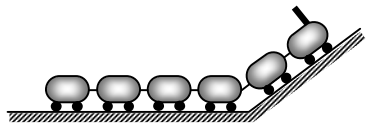


(A) Height of liquid in the tube B is maximum.
(B) Height of liquid in the tubes A and B are same.
(C) Height of liquid in all the three tubes are same.
(D) Height of liquid in the tubes A and C are same.

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A train of length ℓ = 125 m running on a horizontal surface with its engine switched off starts climbing an inclined track of inclination θ = 30° as shown in the figure. Total time spent by train or any part of train on inclination is 5πk sec. The value of k is (g = 10 m/s^2, assume no loss of energy)

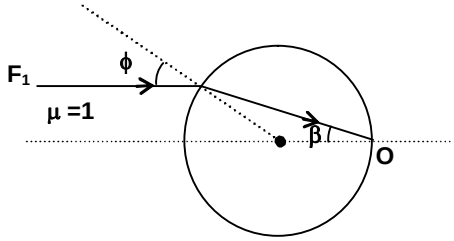


22. We would like to find a relation between mass of black hole and the area of its event horizon A. This area depends on the mass of black hole, speed of light and universal gravitational constant. We write it as $A = G^{\alpha}c^{\beta}m^{\gamma}$. Then the value of $|\alpha + \beta - \gamma|$ is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A ray of light enters a spherical drop of refractive index $\left(\mu = \frac{5}{4}\right)$ as shown in the figure. The value of $100\cos\beta$ is (where ϕ is the angle for which the ray after refraction passes through the point O as shown in the figure.



24. There is a U-tube of different cross sectional area A = 2.5 cm^2 and 4A of limbs fitted with the frictionless pistons each of mass m = 1 kg. An ideal liquid of density ρ = 2000 kg/m^3 is filled in the tube and the tube is accelerated in horizontal direction with a constant acceleration 'a'. Find the value of acceleration 'a' (in m/s^2) for which both pistons stay in the same horizontal level. The length of horizontal section of U-tube is ℓ = 1m. (Take g = 10 m/s^2)

25. A solid sphere is initially kept in open air, pressure exerted by air on it is 10^5 N/m^2. The sphere is lowered into ocean to a depth where pressure is 100 times atmospheric pressure. The fractional change in the volume of the sphere is (Bulk modulus of sphere, B = 4.95×10^8 N/m^2)

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Which of the following statement is correct for the following reaction?

$$CrO_5 + H_2SO_4 \longrightarrow Cr_2(SO_4)_3 + H_2O + O_2$$

(A) It is not a redox reaction.
(B) One mole of CrO_5 will liberate7/4 moles of O_2.
(C) Oxidation number of Cr decreases from +10 to +3 during the reaction.
(D) None of these.

27. A sample of liquid H_2O of mass 18.0 g is injected into an evacuated 7.6 L flask maintained at 27^oC. If vapour pressure of water at 27^oC is 24.63 mm Hg, what weight percentage of the water will be vapourised when the system comes to equilibrium? Assume water vapour behaves as an ideal gas and the volume occupied by water is negligible compared to the volume of the container:
(A) 1%
(B) 10%
(C) 18%
(D) 20%

28. If x is the specific resistance (in S^{-1} cm) of the electrolyte solution and y is the molarity of the solution, then Λ_m (in S cm^2 mol^{-1}) is given by

(A) $\frac{1000x}{y}$

(B) $\frac{1000y}{x}$

(C) $\frac{1000}{xy}$

(D) $\frac{xy}{1000}$

29. Boiling point of which set of aqueous solutions given below will not be identical?
(A) 1 mole NaCl in 1 L water, 2 mole CH_3OH in 1 L water.
(B) 18.4 M H_2SO_4 $\left(d = 1.84 \frac{g}{mL} \text{ and } 98\% \ H_2SO_4 \text{ by mass}\right)$,
500 m H_2SO_4 $\left(d = 1.84 \frac{g}{mL} \text{ and } 98\% \ H_2SO_4 \text{ by mass}\right)$
(C) 0.1 M urea, 0.1 M glucose
(D) 0.11 % by moles (R) – 2-butanol in water, 0.11 % by moles (S) – 2 – butanol in water.

30. Which of the following is a mismatch:
(A) Lyophilic colloids – reversible sols
(B) Associated colloids – micelles
(C) Tyndall effect – Scattering of light by colloidal particles.
(D) Electrophoresis – movement of dispersion medium under the influence of electric field.

31. Which of the following statements is incorrect?
(A) Transition elements exhibit higher enthalpies of atomization as they have stronger interatomic interaction.
(B) IE_2 of $_{23}V < {}_{24}Cr > {}_{25}Mn$ and $_{28}Ni < {}_{29}Cu > {}_{30}Zn$
(C) Ni(II) compounds are more stable than Pt(II) where as Pt(IV) compounds are more stable than Ni(IV).
(D) The elements which give the greatest number of oxidation state does not occur in or near the middle of the transition series.

32. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a:
(A) Hg_2^{+2} salt
(B) Cu^{+2} salt
(C) Ag^{+} salt
(D) Pb^{+2} salt

33. Which of the following compounds does not liberate NH_3 on heating?
(A) $(NH_4)_2SO_4$
(B) $(NH_4)_2CO_3$
(C) NH_4Cl
(D) NH_4NO_3

34. For a certain reaction
$A \longrightarrow$ Products, the $t_{1/2}$ as a function of $[A]_0$ is given as below:

$[A]_0$	:	0.1	0.025
$t_{1/2}$ (min.)	:	100	50

(A) The order is 2.
(B) $t_{1/2}$ would be $100\sqrt{10}$ min. for $[A]_0$ = 1 M.
(C) The order is 1.
(D) $t_{1/2}$ would be 100 min. for $[A]_0$ = 1 M.

35. Which of the following is not an ore of copper?
(A) Azurite
(B) Siderite
(C) Cuprite
(D) Malachite

36. Which of the following pairs of ions is expected to show colour in aqueous solutions?
(A) La^{+3}, Ce^{+3}
(B) La^{+3}, Lu^{+3}
(C) Ce^{+3}, Sm^{+3}
(D) Lu^{+3}, Sm^{+3}

37. The density of steam at 27°C and 8.314×10^4 pascal is 0.8 kg m^{-3}. The compressibility factor would be:
(A) 0.75
(B) 1
(C) 0.88
(D) 1.1

38. The RMS speed of the molecule of diatomic gas is U. When the temperature is doubled, the molecule dissociates into two atoms. The new RMS speed of the atom is:
(A) $\sqrt{2}\,U$
(B) U
(C) 2U
(D) 4U

39. For a perfectly crystalline solid, $C_{P,m} = aT^3$, where a is constant. If $C_{P,m}$ is 0.84 J K^{-1} mol^{-1} at 10 K, molar entropy at 10 K is:
(A) 0.42 J K^{-1} mol^{-1}
(B) 0.28 J K^{-1} mol^{-1}
(C) 4.2 J K^{-1} mol^{-1}
(D) 2.8 J K^{-1} mol^{-1}

40. The ratio of number of σ bond and π bond in Bithionol, which is added to soaps to impart antiseptic properties:
(A) $\frac{26}{3}$
(B) $\frac{13}{3}$
(C) $\frac{19}{3}$
(D) $\frac{22}{3}$

41. How many grams of sulphuric acid is to be dissolved to prepare 200 mL aqueous solution having concentration of H_3O^+ ions = 1 M at 25°C.
(A) 4.9 g
(B) 19.6 g
(C) 9.8 g
(D) 0.98 g

42. Out of Cu^{+2}, Ni^{+2}, Co^{+2} and Mn^{+2}, which ion is precipitated when H_2S is passed through it in the presence of dilute HCl?
(A) Cu^{+2}
(B) Ni^{+2}
(C) Co^{+2}
(D) Mn^{+2}

43. Amongst the following compounds, which one is not responsible for the depletion of ozone layer?
(A) CH_4
(B) $CFCl_3$
(C) NO
(D) Both (A) and (B)

44. 100 mL of 0.05 M $CuSO_4$ (aq.) solution was electrolysed using inert electrodes by passing current till the pH of the resulting solution was 2. The solution after electrolysis was neutralized and then treated with excess KI and formed I_2 was titrated with 0.04 M $Na_2S_2O_3$. Calculate the required volume (in mL) of $Na_2S_2O_3$.
(A) 112.5 mL
(B) 100 mL
(C) 125 mL
(D) 150 mL

45. When an excess of a very dilute aqueous solution of KI is added to a very dilute aqueous solution of $AgNO_3$, the colloidal particles of AgI are associated with which of the following Helmholtz double layer

(A) $AgI \vdots Ag^+ \vdots I^-$

(B) $AgI \vdots K^+ \vdots NO_3^-$

(C) $AgI \vdots I^- \vdots K^+$

(D) $AgI \vdots NO_3^- \vdots Ag^+$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. Among the elements of lanthanoid series $\left({}_{58}Ce - {}_{71}Lu\right)$, the total number of elements which have odd number of electrons in f-orbitals in their ground state configuration.

47. Sucrolose is *x* times sweet in comparison to cane sugar while Aspartame is *y* time sweet in comparison to cane sugar. The ratio $\frac{x}{y}$ is:

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. For first order gaseous reaction, $\log_{10}K$ when plotted against $\frac{1}{T}$ gives a straight line with a slope of – 8000. Calculate the activation energy of the reaction (in Kcal) (Take R = 2 cal k^{-1} mol^{-1})

49. Calculate the concentration of $Na_2S_2O_3.5H_2O$ solution in grams per litre, 10 mL of which just decolourised 15 mL of $\frac{N}{20}$ iodine solution.

50. A fuel cell develops an electrical potential from the combustion of butane at 1 bar at 298 K.
$C_4H_{10}(g) + 6.5O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(\ell)$, $\Delta_r G^0 = -2746$ kJ/mol
What is E^o of the cell (in volt)?

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. Unit vector perpendicular to the vectors $\hat{i}+2\hat{j}-\hat{k}$ and $2\hat{i}+3\hat{j}+\hat{k}$ is

(A) $\frac{1}{\sqrt{35}}\left(5\hat{i}-3\hat{j}-\hat{k}\right)$

(B) $\frac{1}{\sqrt{35}}\left(3\hat{i}-5\hat{j}+\hat{k}\right)$

(C) $\frac{1}{\sqrt{35}}\left(\hat{i}+5\hat{j}-3\hat{k}\right)$

(D) none of these

52. For any three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ (non-zero), $(\vec{a}-\vec{b})\cdot\left[(\vec{b}-\vec{c})\times(\vec{c}-\vec{a})\right]$ is

(A) $2|\vec{c}|$

(B) $3|\vec{b}-\vec{c}|$

(C) $3|\vec{b}-\vec{c}+\vec{a}|$

(D) zero

53. The projection of the line segment P(1, 2, 3) and Q(–1, 4, 2) on the line having direction ratios 2, 3, –6 is

(A) $\frac{7}{8}$

(B) $\frac{6}{7}$

(C) $\frac{8}{7}$

(D) $\frac{9}{2}$

54. Image of the point P(3, 5, 7) in the plane 2x + y + z = 0 is

(A) (9, –1, –1)

(B) (–9, 1, –1)

(C) (–9, –1, 1)

(D) none of these

55. The number of distinct terms in the expansion of $\left(2x+3+\frac{1}{5x}\right)^n$, $n \in N$ is

(A) 3n

(B) 3n + 1

(C) 2n

(D) 2n + 1

56. $\frac{1}{{}^{2n+1}C_5} + \frac{1}{{}^{2n+1}C_6} = \frac{2n+2}{2n+1}\frac{1}{{}^{2n}C_\alpha}$, then α is

(A) 4
(B) 5
(C) 6
(D) none of these

57. Let α, β, γ be distinct complex numbers such that $\frac{\alpha}{1-\beta} = \frac{\beta}{1-\gamma} = \frac{\gamma}{1-\alpha} = \lambda$, then λ is

(A) ω
(B) ω^2
(C) $-\omega^2$
(D) none of these

58. The product of the real part of the roots of $z^2 - z = 5 - 5i$ is

(A) –25
(B) –6
(C) –5
(D) 25

59. The probability that a man can hit a target is $\frac{3}{4}$, he tries 5 times. The probability that he will hit the target at least three times is

(A) $\frac{291}{364}$
(B) $\frac{371}{464}$
(C) $\frac{471}{502}$
(D) $\frac{459}{512}$

60. Let E and F be two independent events such that P(E) > P(F). The probability that both E and F happened is $\frac{1}{12}$ and the probability that neither E nor F happens is $\frac{1}{2}$, then

(A) $P(E) = \frac{1}{3}, P(F) = \frac{1}{4}$
(B) $P(E) = \frac{1}{2}, P(F) = \frac{1}{6}$
(C) $P(E) = \frac{1}{2}, P(F) = \frac{1}{8}$
(D) none of these

61. If a, b, c are all different from zero and $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$, then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is

(A) abc
(B) $\frac{1}{abc}$

(C) $-a - b - c$
(D) -1

62. If, $k \in R^+$, then Det(Adj(λI_4)) is
(A) λ^3
(B) λ^{12}
(C) λ^4
(D) λ

63. A and B are two given matrices such that the order of A is 3×4, if A′B and BA′ are both defined, then
(A) order of B′ is 3×4
(B) order of B′A is 4×4
(C) order of B′A is 3×3
(D) B′A is not defined

64. The sum of integral values of α for which the equation $(x + \alpha)(x + 1991) + 1 = 0$ has integral roots is/are
(A) 1989
(B) 1997
(C) 3982
(D) none of these

65. If for an A.P. $a_1, a_2, a_3, ..., a_n$; $a_1 + a_3 + a_5 = -12$ and $a_1a_2a_3 = 8$, then $a_2 + a_4 + a_6$ is
(A) –12
(B) –16
(C) –18
(D) –2

66. The number of arrangements of the word 'PARABOLA' under the condition that consonants and vowels alternate is
(A) 48
(B) 24
(C) 192
(D) 216

67. The number of ways in which can three people divide among themselves six identical apples, one orange, one lemon, one banana, one pear, one plum and one apricot is
(A) ${}^8C_3$
(B) $(3)^6$
(C) ${}^8C_4 \times (3)^6$
(D) none

68. The mean of 10 numbers is 12.5, the mean of first six numbers is 15 and mean of last five is 10, then sixth number is
(A) 15
(B) 12
(C) 18
(D) 12.5

69. Standard deviation of the first 2n + 1 natural numbers is

(A) $\sqrt{\frac{n(n+1)}{4}}$

(B) $\sqrt{\frac{n(n+1)(2n+1)}{3}}$

(C) $\sqrt{\frac{n(n+1)}{3}}$

(D) $\sqrt{\frac{n(n+1)}{2}}$

70. The coefficients of 3 consecutive terms in the expansion of $(1 + x)^n$ are in the ratio 1 : 7 : 35, then value of 'n' is

(A) 35
(B) 46
(C) 23
(D) none of these

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ be a matrix such that $A^{10} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then the number of factors of $a + b + c + d$ is 'n' where (n – 15) is

72. The equation $x^2 + bx + c = 0$ has distinct roots. If 2 is subtracted from each root, then the result is the reciprocal of original root, then $b^2 + c^2$ is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. At a dance function there are 12 boys and 15 girls. The number of ways in which they can form four couples consisting of different genders is N, then $\frac{N}{10^3}$ is

74. If z is any complex number satisfying $|z – 4 – 3i| = 2$ in argand plane. The maximum and minimum values of |z| are α and β respectively, then $\frac{\alpha+\beta}{4}$ is

75. A purse contains 4 copper and 3 silver coins & another purse contains 6 copper & 2 silver coins. One coin is drawn from any one of these purses. The probability that it is a copper coin is $\frac{\alpha}{\beta}$ ($\alpha, \beta \in N$ & are coprime), then $\frac{\alpha}{\beta+18}$ is